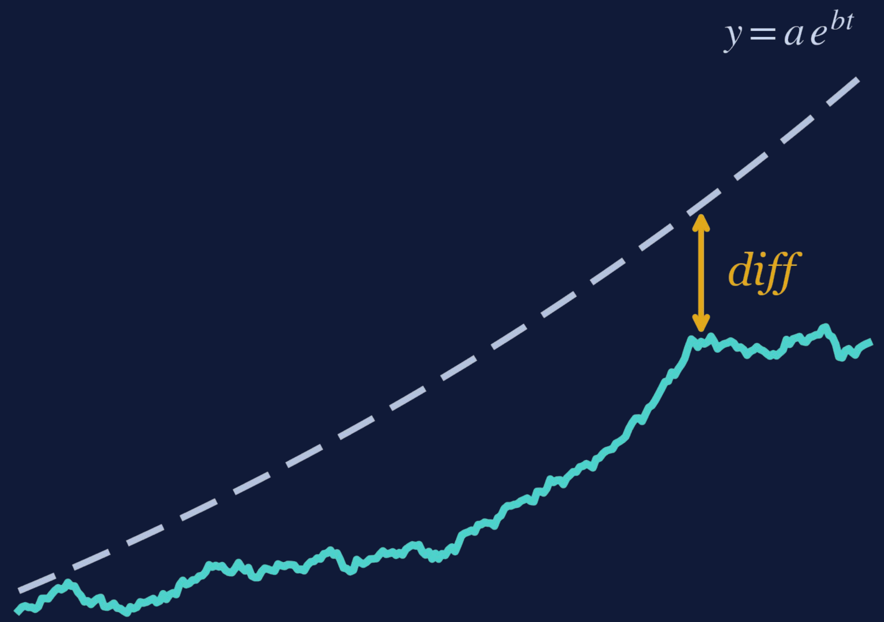
B.SC. RESEARCH PROJECT · SUMMARY REPORT

Risk-Adjusted Performance Analysis Based on Data

Five classic ratios, one new metric — $diff \times r \times b$ — tested across 100 ETFs and 9 market regimes.



Nehorai Sadeh · Aviv Asraf

Supervised by Dr. Dvir Ross · Department of Computer Science, Sami Shamoon College of Engineering, Be'er Sheva · June 2, 2026

ABSTRACT

One new metric against five classics

The project examines the **Sharpe, Treynor, Calmar, Sortino** and **Information** ratios, and proposes a new metric derived from an exponential regression on the asset price.

$$\mathrm{Score} = \mathrm{diff} \times r \times b$$

Across 100 ETFs and 9 market regimes, the new metric acts as a **mean-reversion signal**, while the classic ratios behave as **momentum signals**.

Combining them according to the market regime outperforms each one individually.

diff

Price deviation

Distance of the current price from the trend line

r

Fit quality

Correlation coefficient — quality of fit to the trend

b

Growth slope

Daily growth rate of the fitted trend

THE MOMENTUM FAMILY

The five classic ratios

Each ratio highlights a different aspect of risk — and all five reward strong recent risk-adjusted returns.

Sharpe · 1966

$$\frac{R_p - R_f}{\sigma_p}$$

Excess return per unit of total volatility

Sortino · 1994

$$\frac{R_p - R_f}{\sigma_{\text{down}}}$$

Excess return per unit of downside volatility only

Treynor · 1965

$$\frac{R_p - R_f}{\beta_p}$$

Excess return per unit of systematic risk (β)

Information · 1989

$$\frac{R_p - R_b}{\text{TE}}$$

Consistency of outperformance versus benchmark

Calmar · 1991

$$\frac{R_{\text{ann}}}{|\text{MaxDD}|}$$

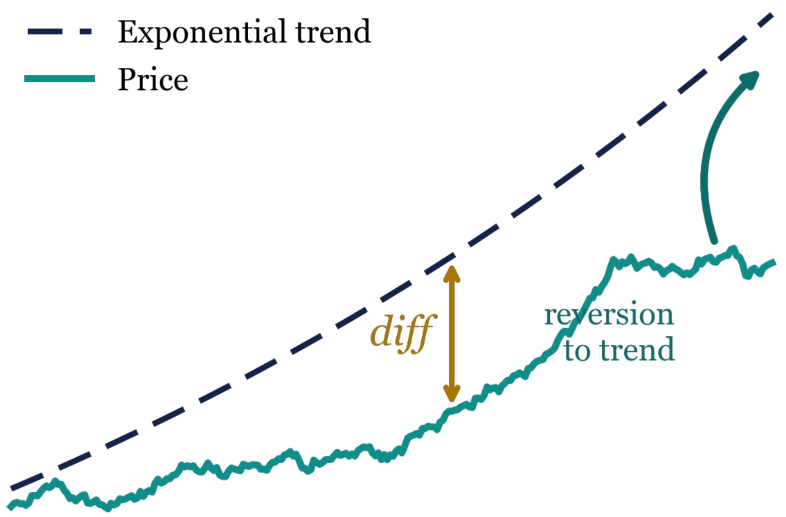
Annual return relative to maximum drawdown

THE NEW METRIC

Score = diff × r × b

We fit an exponential model to the historical price — an OLS regression on log-price:

$$y = a \cdot e^{bt}$$



r

Fit quality

Pearson correlation between log-price and time ($|r| \le 1$)

b

Growth slope

Daily growth rate (log-growth rate)

diff

Price deviation

ExpReg(last) − Price(last): signed distance from the trend

Price below the trend + positive trend → positive score → **predicted UP** (mean reversion).

METHOD

Experiment design

100
exchange-traded funds

180
configurations scanned

9
market regimes

5 × 4
training × test windows

1 — ### Training window

Compute all 6 metrics on each of the 100 funds.

→

2 — ### Test window

Observe direction and realize returns.

→

3 — ### Evaluation

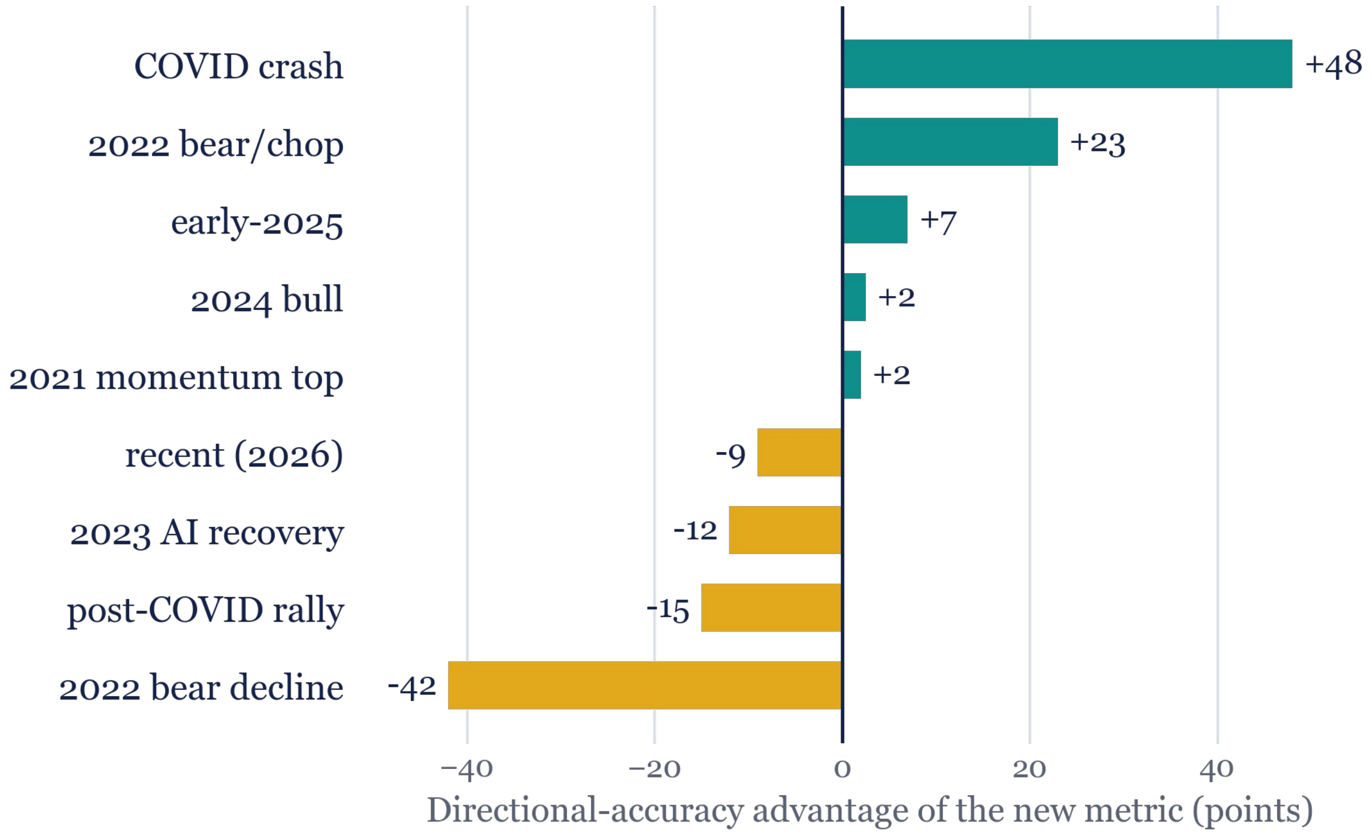
Directional accuracy and long/short portfolio return.

Each metric is converted into a directional forecast: **a positive score predicts an increase, a negative score predicts a decrease**.

In addition, a dollar-neutral long/short portfolio is constructed — 10 assets on each side, weights proportional to |score|, with a strict cap on the short side.

RESULTS

A regime-dependent edge



The new metric wins

Reversals and volatile markets

Classic ratios win

Persistent trends

Key takeaway

The pattern is not random — it is the signature of a mean-reversion signal versus a momentum signal.

CASE STUDIES · THE NEW METRIC WINS

When reversion pays

CASE A · COVID CRASH REBOUND (2020)

96.9% vs **9.2%**

directional accuracy — new metric vs classic ratios

The training window ended near the March 2020 bottom. Almost all ETFs sat below their exponential trend → large positive diff → predicted UP. The classic ratios, depressed by catastrophic returns, forecast continued decline.

Long/short return: +10.9% vs −8.8%

CASE B · 2022 CHOPPY BEAR TAPE (Q4)

90.0% vs **22.6%**

directional accuracy — new metric vs classic ratios

A market with no clear direction — mean-reverting. Extended names below trend snapped back; names above it faded. The advantage is not unique to a single crash.

Long/short return: +8.1% vs +2.4%

CASE STUDIES · THE CLASSIC RATIOS WIN

When the trend persists

CASE C · POST-COVID MOMENTUM RALLY

2.0% vs **93.9%**

directional accuracy — new metric vs classic ratios

End of 2020 — the market had risen for months. Most ETFs were above their trend → negative diff → predicted a pullback that never came. Momentum continued; the classic ratios were right in almost every case.

CASE D · 2022 SUSTAINED DECLINE

8.1% vs **79.0%**

directional accuracy — new metric vs classic ratios

A prolonged, directional decline. Names below trend kept falling — the mean-reversion signal “caught a falling knife.”

Long/short return: −6.4% vs +7.8%

When the trend persists, mean-reversion forecasting is systematically wrong — the mirror image of Case A.

DISCUSSION

Two complementary signal types

The new metric

MEAN REVERSION

Signal positive when price sits below a stable trend

Predicts a return to the trend

Wins in reversals and volatile markets (Cases A, B)

The classic ratios

MOMENTUM

Signal large after strong risk-adjusted returns

Predicts continuation of the move

Wins in sustained trends (Cases C, D)

The literature (Jegadeesh & Titman, 1993; Daniel & Moskowitz, 2016) documents “momentum crashes” at reversals — Case A is a textbook example.

Takeaway: a regime-aware combination — switching between the two signal types according to volatility, trend strength, or the size of diff — is expected to outperform either family on its own.

Complementary, not competing

1. **diff × r × b is a mean-reversion signal** — not a replacement for the classic ratios.
2. **The classic ratios behave as momentum signals** — direct empirical evidence.
3. **The two families are complementary** — each wins where the other fails.
4. **Regime-based combination** outperforms any single metric on its own.

NEXT STEP

Walk-forward backtest

Roll the training window day by day, recalculating metrics and rebalancing portfolios into a continuous equity curve — testing:

- whether the regime-dependent edge survives sequential rebalancing
- regime filters (volatility, trend strength, |diff|) cycle by cycle
- transaction costs, turnover, and short-borrow constraints